University of BRISTOL
Centre for Academic Language and Development

Foundations of Computer Programming - LANG00052

Week 03

Pseudocode

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Lecture Outline

- Understand what pseudocode is and why it’s important.
- Learn the fundamental structure of pseudocode
- Understand how to represent decision-making and repetition in pseudocode.
- Ensure clarity and effectiveness in writing pseudocode
- Identify and correct common errors in pseudocode writing
- Discussion and clarify doubts

WHAT IS PSEUDOCODE ?

Pseudo is from the Greek ‘psevdes’ ,meaning false

Pseudocode is not an actual programming language

It therefore doesn’t have the strict syntax normal language have

PSEUDOCODE

Pseudocode is structured English for describing algorithms concisely. It is made up of two words, namely, pseudo meaning imitation and code meaning instructions.

As the name suggests, pseudocode does not obey the syntax rules of any programming language i.e. it is not a real programming code. It allows the designer to focus on main logic without being distracted by programming languages syntax.

PYTHON CODE

```python

class Main():
    def __init__(self,name):
        self.name = name

    def myMethod(self):
        print(self.name)

name =Main("I just got executed!")

name.myMethod()

```

OUTPUT

```
I just got executed!
```

JAVA CODE

```java

class Main {
    static void myMethod() {
      System.out.println("I just got executed!");
    }
    Run | Debug
    public static void main(String[] args) {
    myMethod();
  }
}
```

OUTPUT

```
dcd96cd7\bin' 'Main'
I just got executed!
```

COMMON PSEUDOCODE NOTATION

There is no strict set of standard notions for pseudocode, but some of the most widely recognised are:

START/END - Indicate the beginning and end of the pseudocode.(BEGIN or START & END or STOP)

- INPUT –indicates a user will be inputting something
- OUTPUT- indicates that an output will appear on the screen
- WHILE- a loop(iteration that has a condition at the beginning)
- FOR- a counting loop(iteration)
- REPEAR- UNTIL –a loop (iteration) that has a condition at the end
- IF -THEN- ELSE – a decision (selection) in which a choice is made

Common Keywords

BEGIN, END
IF, THEN, ELSE, ENDIF
FOR, TO, DO, ENDFOR
WHILE, ENDWHILE
REPEAT, UNTIL
INPUT, OUTPUT
SET, PRINT, RETURN

WRITE A PSEUDOCODE TO FIND THE SUM OF TWO NUMBERS.

```
begin
    numeric nNum1,nNum2,nSum
    display "ENTER THE FIRST NUMBER : "
    accept nNum1
    display "ENTER THE SECOND NUMBER : "
    accept nNum2
    compute nSum=nNum1+nNum2
    display "SUM OF THESE NUMBER : " nSum
end
```

ADVANTAGES OF PSEUDOCODE

1. It allows the designer to focus on main logic without being distracted by programming languages syntax.
2. Since it is language independent, it can be translated to any computer language code.
3. It allows designer to express logic in plain natural language
4. It is easier to write actual code using pseudocode.
5. Unlike algorithms, pseudocodes are concise, so pseudocodes are more readable and easier to modify.

DISDVANTAGES OF PSEUDOCODE

1. There are no accepted standards for writing pseudocodes and designer use their own style while writing pseudocodes.
2. Pseudocode cannot be compiled and executed so its correctness cannot be verified by using computers.

```
BEGIN IF
   INPUT age
   IF age >= 18 THEN
      PRINT "You are eligible to vote."
   FOR
      PRINT "You are not eligible to vote."
   ENDIF
   SET sum - 0
   FOR i - 1 TO 10 DO
      sum - sum + I
   END IF
   ENDFOR
   PRINT "The sum is:", sum
   FUNCTION AddNumbers (a, b) RETURNS integer
      RETURN a + b
   END FUNCTION
   SET result ← AddNumbers(5, 3)
   PRINT "The result of addition is:", result
END IF
```

```
BEGIN
   INPUT age
   IF age >= 18 THEN
      PRINT "You are eligible to vote."
   ELSE
      PRINT "You are not eligible to vote."
   END IF
   SET sum - 0
   FOR i - 1 TO 10 DO
      sum - sum + i
   END FOR
   PRINT "The sum is:", sum
   FUNCTION AddNumbers(a, b) RETURNS integer
      RETURN a + b
   ENDFUNCTION
   SET result ← AddNumbers(5, 3)
   ELSE  "The result of addition is:", result
END
```

Best Practices for Writing Pseudocode

- Keep it Simple:

Use plain language and avoid complex structures.

- Be Consistent:

Stick to a uniform format throughout.

- Focus on Logic:

Concentrate on the algorithm's flow rather than language-specific syntax.

- Use Comments:

Explain steps and complex logic where necessary.

Common errors in pseudocode writing

- Overcomplicating the pseudocode.
- Ignoring logical flow and sequence.
- Mixing programming language syntax with pseudocode.

Arithmetic operators	
Symbol	**Description**
+	Add
-	Subtract
/	Divide
*	Multiply
^	Exponent
MOD	Modulo
DIV	Integer division

Relational operators

Symbol	Description
=	equal to
<>	not equal to
>	greater than
>=	greater than or equal to
<	less than
<=	less than or equal to

Logical operators	
Symbol	**Description**
AND	Returns true if both conditions are true.
OR	Returns true if any of the conditions are true.
NOT	Reverses the outcome of the expression; true becomes false, false becomes true.

Study the next 4 slides before the fun begins!

Input/output		
Syntax	Explanation of syntax	Example
SEND <expression> TO DISPLAY	Sends output to the screen.	SEND ‘Have a good day.’ TO DISPLAY
RECEIVE <identifier> FROM (type) <device>	Reads input of specified type.	RECEIVE Name FROM (STRING) KEYBOARD
		RECEIVE LengthOfJourney FROM (INTEGER) CARD_READER
		RECEIVE YesNo FROM (CHARACTER) CARD_READER

Repetition

Syntax	Explanation of syntax	Example
WHILE <condition> DO **<command> END WHILE**	Pre-conditioned loop. Executes <command> while <condition> is true.	**WHILE Flag = 0 DO** **SEND 'All well' TO DISPLAY END WHILE**
REPEAT **<command> UNTIL** **<expression>**	Post-conditioned loop. Executes <command> until <condition> is true. The loop must execute at least once.	**REPEAT** **SET Go TO Go + 1 UNTIL Go** **= 10**
REPEAT <expression> TIMES **<command> END REPEAT**	Count controlled loop. The number of times <command> is executed is determined by the expression.	**REPEAT 100-Number TIMES** **SEND '*' TO DISPLAY END REPEAT**
FOR <id> FROM **<expression> TO** **<expression> DO** **<command> END FOR**	Count controlled loop. Executes <command> a fixed number of times.	**FOR Index FROM 1 TO 10 DO** **SEND ArrayNumbers [Index] TO DISPLAY END FOR**
FOR <id> FROM **<expression> TO** **<expression> STEP** **<expression> DO** **<command> END FOR**	Count controlled loop using a step.	**FOR Index FROM 1 TO 500** **STEP 25 DO** **SEND Index TO DISPLAY END FOR**
FOR EACH <id> FROM **<expression> DO** **<command> END FOREACH**	**Count controlled loop. Executes for each element of an array.**	**SET WordsArray TO ['The', 'Sky', 'is', 'grey']** **SET Sentence to "** **FOR EACH Word FROM** **WordsArray DO SET Sentence TO** **Sentence & Word & " END FOREACH**

Selection		
Syntax	**Explanation of syntax**	**Example**
IF <expression> THEN <command> END IF	If <expression> is true then command is executed.	IF Answer = 10 THEN SET Score TO Score + 1 END IF
IF <expression> THEN <command> ELSE <command> END IF	If <expression> is true then first <command> is executed, otherwise second <command> is executed.	IF Answer = ‘correct’ THEN SEND ‘Well done’ TO DISPLAY ELSE SEND ‘Try again’ TO DISPLAY END IF

Variables and arrays		
Syntax	**Explanation of syntax**	**Example**
SET Variable TO <value>	Assigns a value to a variable.	SET Counter TO 0 SET MyString TO ‘Hello world’
SET Variable TO <expression>	Computes the value of an expression and assigns to a variable.	SET Sum TO Score + 10 SET Size to LENGTH (Word)
SET Array [index] TO <value>	Assigns a value to an element of a one- dimensional array.	SET ArrayClass [1] TO ‘Ann’ SET ArrayMarks [3] TO 56
SET Array TO [<value>, …]	Initialises a one- dimensional array with a set of values.	SET ArrayValues TO [1, 2, 3, 4, 5]
SET Array [RowIndex, ColumnIndex] TO <value>	Assigns a value to an element of a two- dimensional array.	SET ArrayClassMarks [2,4] TO 92

Question 1

WRITE A PSEUDOCODE TO FIND THE LARGEST OF TWO NUMBERS.

Question 2

WRITE A PSEUDOCODE TO FIND THE AREA OF RECTANGLE.

Question 3

WRITE A PSEUDOCODE TO FIND THE CIRCUMFERENCE OF CIRCLE.

Question 4

WRITE A PSEUDOCODE TO FIND THE AREA OF TRIANGLE.

Time for questions